

Adaptive Shared Control for Collaborative Robots via Vision-Based Motion Prediction and Ergonomic Evaluation

Abstract—As human-robot collaboration (HRC) becomes increasingly prevalent in dynamic environments, seamlessly blending human intent with autonomous robotic execution remains a critical challenge. Standard motion prediction models often exhibit overconfidence or fail to account for human ergonomic comfort, leading to jerky robotic responses and operator fatigue. This paper proposes a novel adaptive shared control framework designed for collaborative robot arms. By integrating a deep learning-based motion predictor with an adaptive weight generator, the system explicitly fuses the reliability of algorithmic predictions with real-time human arm comfort scores. A trajectory smoother dynamically allocates control authority, ensuring stable and fluent cooperation. The proposed control loop, implemented via a quaternion-based inverse-kinematics formulation and an optimal LQR controller, strictly constrains absolute joint positions to ensure robust tracking. Experimental validation on a physical collaborative robot demonstrates significant improvements in task fluency, reduced control effort, and enhanced human postural comfort compared to traditional fixed-authority baselines.

Index Terms—Human-Robot Collaboration, Shared Control, Deep Learning, Motion Prediction, Quaternions.

I. INTRODUCTION

THE integration of collaborative robots (cobots) into shared workspaces requires advanced control paradigms that can seamlessly adapt to human behavior. While vision-based skeleton tracking and deep learning (DL) models have significantly improved human motion prediction, relying solely on autonomous predictions can be dangerous if the algorithm's confidence drops due to occlusion or sudden movements.

Existing literature often overlooks the necessity of dynamically blending robot autonomy with human ergonomic requirements. To address this gap, this paper introduces an adaptive shared control architecture. The key contributions of this work are threefold:

- 1) A novel adaptive weight generation mechanism that fuses DL prediction reliability with real-time arm comfort scores mapped against an optimal human pose dataset.
- 2) A robust trajectory smoothing filter that ensures stable control authority blending without inducing oscillatory behavior in the cobot.
- 3) A rigorously constrained, quaternion-based control formulation validated on a physical robot platform, demonstrating statistical superiority over fixed-weight blending methods.

II. RELATED WORK

A. Human Motion Prediction in HRC

To enable proactive robot behaviors in collaborative workspaces, early and accurate estimation of human motion trajectories is critical, as highlighted by several comprehensive overviews [1], [2]. Traditional deterministic architectures have been widely utilized to capture temporal dependencies in human motion sequences due to their high computational efficiency. Specifically, Recurrent Neural Networks (RNNs) have been successfully applied to trajectory prediction in collaborative assembly and intention estimation [3]–[5]. To address the vanishing gradient problem in long sequences, Long Short-Term Memory (LSTM) networks have been adopted for dynamic co-carrying and intent recognition tasks [6]–[9]. Furthermore, Gated Recurrent Units (GRUs) offer similar sequence modeling capabilities with a simplified architecture, making them highly efficient for real-time motion prediction [10], [11]. However, these point-prediction models inherently lack the ability to represent model uncertainty. To address this, probabilistic approaches such as Gaussian Process Regression (GPR) have been introduced to forecast future movements while providing continuous uncertainty bounds for safe planning [12]. Furthermore, advanced architectures exploit the physical configuration of the human body using Spatial-Temporal Graph Convolutional Networks (STGCN) to improve robustness against sensor noise and occlusions [13], while others estimate uncertainty via Monte Carlo Dropout in Graph Neural Networks [14]. Hierarchical models have also been developed to bridge symbolic task recognition with continuous trajectory regression [15], [16].

While probabilistic models and complex graphs offer rich uncertainty profiles and robustness, they often impose significant computational overhead, which can induce latency in high-frequency real-time control loops. Deep learning models based on sequence forecasting have proven highly effective for trajectory forecasting without the need for overly complex architectures [17]. Our framework balances this by employing a streamlined Deep-GRU network that guarantees low inference latency, while relying on a dedicated, real-time prediction reliability metric to evaluate prediction confidence dynamically, rather than computing computationally heavy probabilistic bounds.

B. Ergonomic Assessment in HRC

Incorporating ergonomics into HRC is key to mitigating work-related musculoskeletal disorders (WMSDs). Early efforts focused on digitizing standardized observational tools,

such as the Rapid Upper Limb Assessment (RULA) [18], [19] and the Rapid Entire Body Assessment (REBA) [20], utilizing contactless vision-based trackers to monitor operator posture. To overcome the limitations of static postural indices, researchers have developed dynamic biomechanical models that estimate joint overloading torques [21] and cumulative kinematic wear to dynamically delegate high-risk procedures to the robot [22]. Concurrently, robotic response strategies have evolved from static workstation reconfigurations [23] to active haptic guidance, where the cobot generates virtual ergonomic forces to proactively guide the human spine toward healthier configurations [24].

Although dynamic torque models provide highly personalized biomechanical assessments, they often require intrusive wearable sensors (e.g., IMUs or sEMG) that are impractical in industrial settings. Observational methods offer non-intrusive alternatives but are typically discrete and non-differentiable. To achieve continuous, contactless evaluation, our approach models optimal RULA-compliant postures as a multidimensional Gaussian distribution. By calculating the Mahalanobis distance from this optimal distribution, the system generates a smooth, real-time arm comfort score relying exclusively on standard RGB-D cameras without requiring physical sensors on the worker.

C. Authority Blending and Shared Control

Establishing an effective shared control paradigm requires robust methodologies to smoothly allocate control authority between the human operator and the autonomous robot. Traditional approaches utilize state-dependent linear blending formalisms, where the authority weight is governed directly by prediction confidence [25], [26]. To prevent the human from physically fighting against overconfident robotic assistance, minimum intervention shared control frameworks using Deep Reinforcement Learning (DRL) have been proposed to adaptively constrain support [27], [28]. Alternatively, implicit blending through Dynamic Movement Primitives (DMPs) treats human inputs as external disturbances, bypassing explicit algebraic weights [29]. Recent paradigms have also explored differential cooperative game theory (DCGT) for force-based role adaptation [30] and generative co-policies based on diffusion models to plan joint human-robot action sequences [31].

While game-theoretic and generative models exhibit highly sophisticated negotiation behaviors, they often lack straightforward stability guarantees and require exhaustive interaction datasets or complex force-torque sensors at the robot joints. In contrast, our framework introduces a novel Product-of-Experts (PoE) fusion mechanism governed by a Lyapunov-stable update law. This approach ensures mathematical stability and completely eliminates the erratic authority chattering often observed in standard linear blending.

D. Summary and Research Gaps

Despite significant advancements, a noticeable gap remains in simultaneously integrating algorithmic reliability and physiological state into the robotic control loop. Most predictive

models lack efficient online reliability metrics, and ergonomic evaluations are rarely synthesized directly into the low-level authority arbiter. Furthermore, many shared control schemes suffer from discontinuous weight transitions or require expensive hardware to guarantee stable interaction. Our framework addresses these limitations by establishing a closed-loop system that unifies Deep-GRU motion prediction, continuous vision-based Mahalanobis comfort scoring, and a mathematically stable authority blending law, thereby optimizing both performance and human well-being.

III. PROPOSED METHODOLOGY

This section presents the mathematical formulation of the proposed adaptive shared control framework for human-robot co-carrying. The system operates in a real-time control loop, processing inputs from an RGB-D camera and commanding a collaborative robot arm. The overall architecture is illustrated in Fig. 1.

The framework consists of six interconnected modules that execute sequentially within each control tick:

- 1) The **vision pipeline** extracts 3D skeletal landmarks from the RGB-D stream, providing the human wrist position $\mathbf{p}_{wr}(t)$ and the arm joint angle vector $\Theta_{arm}(t)$.
- 2) The **Deep-GRU predictor** takes a sliding window of historical wrist positions as input and outputs the predicted future wrist position $\hat{\mathbf{p}}_{pre}(t+k)$ for a horizon of k steps.
- 3) The **Adaptive Weight Generator** evaluates two real-time metrics—prediction reliability s_r and arm comfort s_e —and synthesizes a dynamic blending weight $w(t) \in [0, 1]$.
- 4) The **Trajectory Smoother** fuses the predicted trajectory with the robot's current end-effector feedback $\mathbf{p}_{fb}(t)$ using $w(t)$, producing a velocity-constrained Cartesian command $\mathbf{p}_{sm}(t)$.
- 5) The **DLS Inverse Kinematics** solver resolves $\mathbf{p}_{sm}(t)$ into a joint reference configuration $\mathbf{q}_{ref}(t) \in \mathbb{R}^6$.
- 6) The **LQR Velocity Controller** generates optimal joint velocities $\dot{\mathbf{q}}_{cmd}(t)$ and positions $\mathbf{q}_{cmd}(t)$ sent to the robot controller.

The feedback loop is closed by the robot's joint state publisher, which provides real-time joint positions $\mathbf{q}_{fb}$ and velocities $\dot{\mathbf{q}}_{fb}$, as well as the forward kinematics solution $\mathbf{p}_{fb} = FK(\mathbf{q}_{fb})$. The remainder of this section details each module mathematically.

A. Vision-based Motion Prediction

1) **Deep-GRU Architecture and Forward Pass:** The motion prediction module employs a stacked Deep Gated Recurrent Unit (GRU) architecture. GRUs are selected over standard LSTMs because their simplified gating mechanism—combining the forget and input gates into a single update gate—reduces the parameter count while retaining comparable sequence modeling capacity [32]. This reduction is particularly advantageous in real-time co-carrying, where inference must complete within a single control tick.

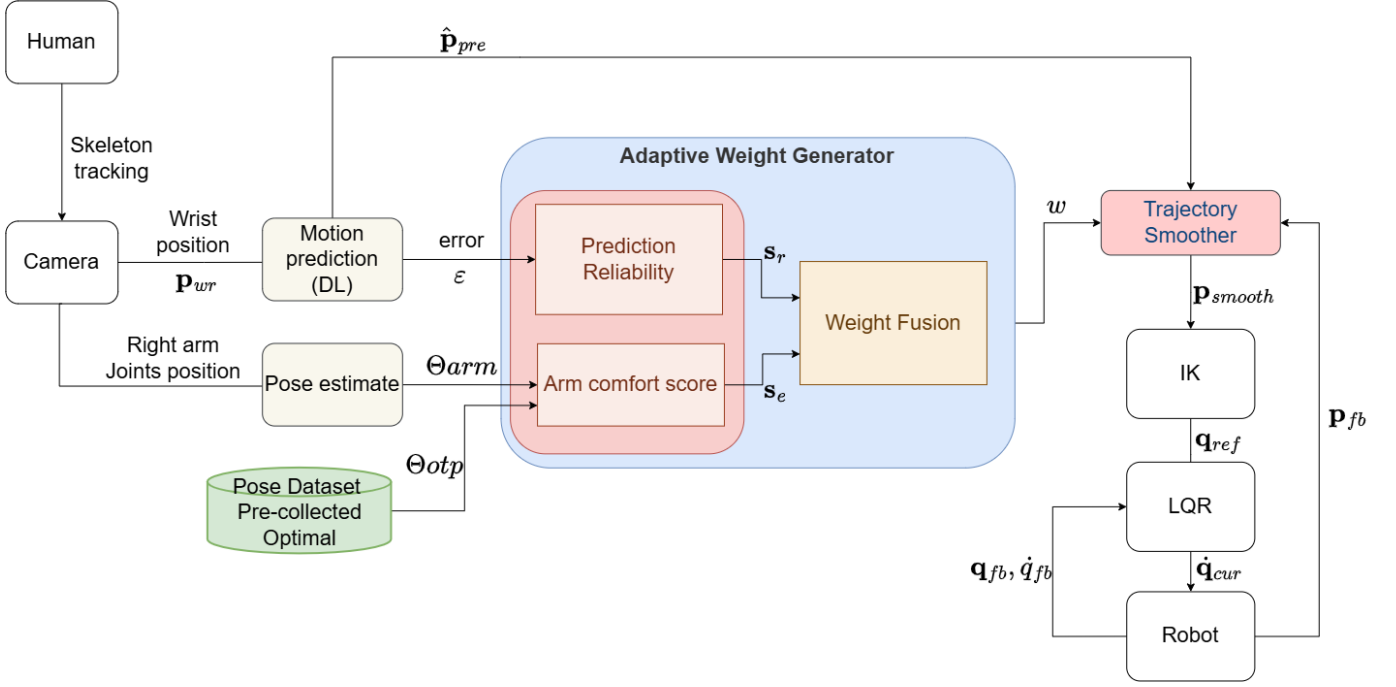


Fig. 1. Proposed adaptive shared control framework. The vision pipeline extracts the human wrist position $\mathbf{p}_{wr}$ and arm joint angles Θ_{arm} . The Adaptive Weight Generator fuses the prediction reliability score s_r and the arm comfort score s_e to produce the blending weight w . The Trajectory Smoother, IK solver, and LQR controller form the robot execution chain.

To construct a state representation suitable for temporal sequence modeling, both the instantaneous position and its first-order derivative are required. Position alone is insufficient because two identical hand positions may correspond to very different future trajectories depending on the current velocity. Accordingly, at each time step t the state of the human hand is captured by a 6-dimensional feature vector comprising 3D position and velocity:

$$\phi_t = [x_t, y_t, z_t, \dot{x}_t, \dot{y}_t, \dot{z}_t]^T \in \mathbb{R}^6 \quad (1)$$

where (x_t, y_t, z_t) are the Cartesian coordinates of the wrist landmark in the camera frame and $(\dot{x}_t, \dot{y}_t, \dot{z}_t)$ are the corresponding finite-difference velocities. This choice provides a first-order Markov representation: given the current position and velocity, future hand motion can be estimated without requiring higher-order derivatives, which would amplify sensor noise.

Rather than feeding a single frame to the network, a sliding window of N consecutive observations is used to provide sufficient temporal context for capturing motion trends such as acceleration and curvature changes. The input sequence is:

$$\Phi_t = [\phi_{t-(N-1)}, \phi_{t-(N-2)}, \dots, \phi_t] \in \mathbb{R}^{N \times 6} \quad (2)$$

This matrix stacks the N most recent feature vectors in chronological order, enabling the recurrent layers to learn how the trajectory evolves over the observation horizon.

Because the six channels of ϕ_t span different physical scales (positions in meters, velocities in meters per second), direct concatenation would bias the network toward whichever channel has the largest numeric range. Before entering the

GRU layers, Φ_t is therefore normalized channel-wise using a Min-Max Scaler:

$$\phi'_{t,j} = \frac{\phi_{t,j} - \phi_{j,\min}}{\phi_{j,\max} - \phi_{j,\min}}, \quad j = 1, \dots, 6 \quad (3)$$

where $\phi_{j,\min}$ and $\phi_{j,\max}$ are the minimum and maximum values of the j -th channel, respectively, computed from the training dataset. After normalization each channel lies in $[0, 1]$, ensuring balanced gradient flow across all input dimensions.

The structural flow of the proposed network is illustrated in Fig. 2. The architecture consists of L stacked GRU layers, each with d_h hidden units, followed by a fully connected output layer. Stacking multiple recurrent layers enables the network to form hierarchical temporal abstractions: lower layers capture short-range patterns (e.g., tremor and jitter) while upper layers encode longer-range motion intent (e.g., reaching and placing). All hidden states are initialized to zero: $\mathbf{h}_0^{(l)} = \mathbf{0}$ for all $l = 1, \dots, L$.

Within each GRU cell, two gating mechanisms govern how much of the past hidden state is retained versus overwritten. For the first GRU layer ($l = 1$), which receives the normalized input ϕ'_t directly, the learnable weight matrices $\mathbf{W}_z^{(1)}, \mathbf{W}_r^{(1)}, \mathbf{W}_h^{(1)} \in \mathbb{R}^{d_h \times (d_h+6)}$ and bias vectors $\mathbf{b}_z^{(1)}, \mathbf{b}_r^{(1)}, \mathbf{b}_h^{(1)} \in \mathbb{R}^{d_h}$ parameterize the update gate $\mathbf{z}_t^{(1)}$, reset gate $\mathbf{r}_t^{(1)}$, candidate hidden state $\tilde{\mathbf{h}}_t^{(1)}$, and final hidden state $\mathbf{h}_t^{(1)}$ as follows:

$$\mathbf{z}_t^{(1)} = \sigma(\mathbf{W}_z^{(1)}[\mathbf{h}_{t-1}^{(1)}; \phi'_t] + \mathbf{b}_z^{(1)}) \quad (4)$$

$$\mathbf{r}_t^{(1)} = \sigma(\mathbf{W}_r^{(1)}[\mathbf{h}_{t-1}^{(1)}; \phi'_t] + \mathbf{b}_r^{(1)}) \quad (5)$$

$$\tilde{\mathbf{h}}_t^{(1)} = \tanh(\mathbf{W}_h^{(1)}[\mathbf{r}_t^{(1)} \odot \mathbf{h}_{t-1}^{(1)}; \phi'_t] + \mathbf{b}_h^{(1)}) \quad (6)$$

$$\mathbf{h}_t^{(1)} = (1 - \mathbf{z}_t^{(1)}) \odot \mathbf{h}_{t-1}^{(1)} + \mathbf{z}_t^{(1)} \odot \tilde{\mathbf{h}}_t^{(1)} \quad (7)$$

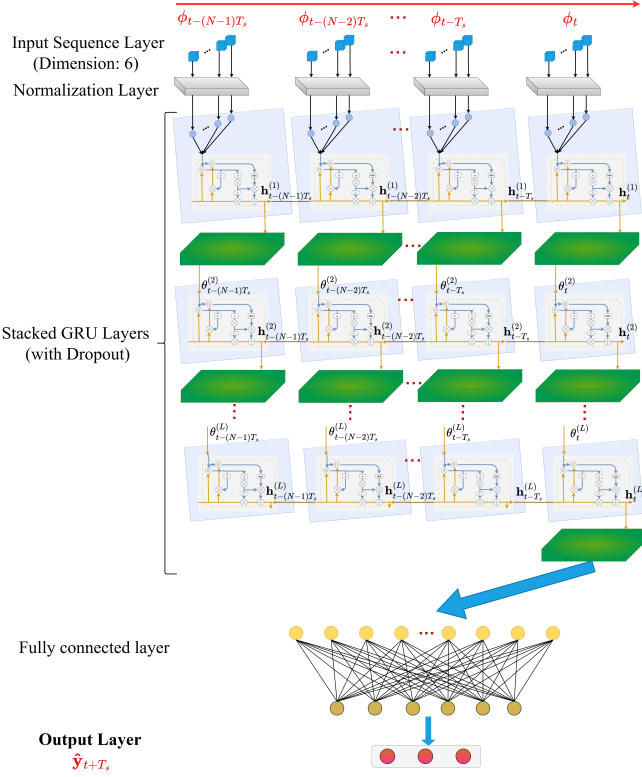


Fig. 2. Proposed Deep-GRU neural network architecture for 3D trajectory prediction. The model processes a sequence of N normalized feature vectors through L stacked GRU layers with dropout regularization, mapping the final hidden state to the predicted future position via a dense output layer.

where $\sigma(\cdot)$ is the sigmoid function, $[\cdot; \cdot]$ denotes vertical concatenation, and $\odot$ denotes the Hadamard (element-wise) product. Physically, the update gate z_t determines the proportion of the new candidate state that replaces the memory from the previous time step: when $z_t \rightarrow 1$ the cell discards old information and adopts the new observation; when $z_t \rightarrow 0$ it retains the existing hidden state unchanged. The reset gate r_t controls how much of the previous hidden state is exposed when computing the candidate, allowing the cell to “forget” irrelevant history.

For intermediate layers ($l = 2, \dots, L$), the input is replaced by the dropout-regularized output of the previous layer $\theta_t^{(l-1)} = \mathbf{m}_t^{(l-1)} \odot \mathbf{h}_t^{(l-1)}$, where $\mathbf{m}_t^{(l-1)} \sim \text{Bernoulli}(1 - p_{drop})$ during training and $\mathbf{m}_t^{(l-1)} = 1$ during inference to ensure deterministic control outputs. Dropout between layers serves as an implicit regularizer, preventing the network from relying excessively on any single hidden unit and thus improving generalization to unseen co-carrying scenarios.

After the entire N -step input sequence Φ_t has been processed, the final hidden state $\mathbf{h}_N^{(L)}$ of the deepest GRU layer encodes a compressed summary of the observed trajectory. This summary is decoded into a spatial prediction by a fully connected output layer:

$$\hat{\mathbf{p}}_{pre}(t+k) = \mathbf{W}_y \cdot \mathbf{h}_N^{(L)} + \mathbf{b}_y \quad (8)$$

where $\mathbf{W}_y \in \mathbb{R}^{3 \times d_h}$ is the output weight matrix, $\mathbf{b}_y \in \mathbb{R}^3$ is the output bias, and k denotes the prediction horizon in

discrete time steps. The output $\hat{\mathbf{p}}_{pre}(t+k) \in \mathbb{R}^3$ represents the predicted 3D Cartesian wrist position at the future time instant $t+k$, providing the downstream controller with anticipatory information about where the human intends to move.

2) *Training*: The network is trained offline on a dataset of recorded co-carrying trials, where ground-truth future wrist positions $\mathbf{y}_i \in \mathbb{R}^3$ are available for each training sample i . The objective is to minimize the Mean Squared Error (MSE) loss, which penalizes the squared Euclidean distance between the predicted and actual future positions:

$$\mathcal{L} = \frac{1}{M} \sum_{i=1}^M \|\mathbf{y}_i - \hat{\mathbf{p}}_{pre,i}\|^2 \quad (9)$$

where M is the total number of training samples. Minimizing this loss drives the network to produce spatially accurate trajectory forecasts in a least-squares sense.

Optimization is performed using the Adam algorithm [33], which adapts the learning rate for each parameter individually based on estimates of the first and second moments of the gradient. The network parameters Θ (comprising all weight matrices and biases across the L GRU layers and the output layer) are iteratively updated via:

$$\Theta \leftarrow \Theta - \eta_{adam} \frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t + \xi}} \quad (10)$$

Here, η_{adam} denotes the learning rate, $\hat{\mathbf{m}}_t$ and $\hat{\mathbf{v}}_t$ are the bias-corrected first and second moment estimates of the gradients respectively, and ξ is a small scalar constant to prevent division by zero.

3) *Real-time Prediction Error*: Once the trained network is deployed in the online control loop, a mechanism is needed to evaluate whether its predictions remain trustworthy under the current operating conditions. Prediction quality may degrade at runtime due to sudden changes in operator behavior, sensor occlusions, or distribution shifts between the training and deployment environments.

To quantify this, the system compares the actual wrist position $\mathbf{p}_{wr}(t)$ observed at the current time step t against the prediction $\hat{\mathbf{p}}_{pre}(t|t-k)$ that was generated k steps earlier for the present instant:

$$\epsilon(t) = \|\hat{\mathbf{p}}_{pre}(t|t-k) - \mathbf{p}_{wr}(t)\|_2 \quad (11)$$

It is important to note that this error formulation compares the historical prediction made for the present instant against the actual observation at that same instant. This avoids the common pitfall of comparing a prediction made at time t for time $t+k$ against the position measured at time t , which would conflate genuine prediction error with the natural displacement due to the operator’s intended motion over the horizon k . The resulting scalar $\epsilon(t)$ therefore accurately reflects the real-time model accuracy and serves as the primary input to the Prediction Reliability module described in Section III-C.

B. Human Joint Angle Estimation

Beyond predicting where the operator’s hand will move, the framework also monitors *how* the operator is moving, specifically whether the current arm posture is ergonomically

sound. Prolonged co-carrying in awkward configurations can lead to work-related musculoskeletal disorders (WMSDs), particularly in the shoulder, elbow, and wrist joints. To quantify postural risk, five biomechanically relevant arm joint angles are extracted in real time from the same 3D skeletal landmarks that feed the motion predictor.

Let the 3D position vectors of the neck (mid-shoulder), torso (mid-hip), right shoulder, right elbow, right wrist, and right index finger be denoted $\mathbf{r}_n$, $\mathbf{r}_t$, $\mathbf{r}_s$, $\mathbf{r}_e$, $\mathbf{r}_w$, and $\mathbf{r}_i$, respectively. The neck and torso reference points are computed as the midpoints of their bilateral counterparts to define a body-centered vertical axis:

$$\mathbf{r}_n = \frac{1}{2}(\mathbf{r}_{L\text{-shoulder}} + \mathbf{r}_{R\text{-shoulder}}) \quad (12)$$

$$\mathbf{r}_t = \frac{1}{2}(\mathbf{r}_{L\text{-hip}} + \mathbf{r}_{R\text{-hip}}) \quad (13)$$

The five angles are chosen to correspond to the anatomical degrees of freedom evaluated by the Rapid Upper Limb Assessment (RULA) methodology [18], covering the major risk factors for upper-limb WMSDs: upper arm flexion/extension, upper arm abduction, forearm flexion, forearm pronation/supination, and wrist deviation. Together, they form a compact yet ergonomically meaningful representation encoded as a 5-dimensional angle vector:

$$\Theta_{arm}(t) = [\alpha_s, \alpha_c, \beta_s, \beta_t, \gamma_s]^\top \in \mathbb{R}^5 \quad (14)$$

Each component is computed via the dot-product formula $\angle(\mathbf{a}, \mathbf{b}) = \arccos(\mathbf{a}^\top \mathbf{b} / (\|\mathbf{a}\| \|\mathbf{b}\|))$ applied to the appropriate anatomical vectors, as detailed below.

The **upper arm sagittal angle** α_s measures the elevation of the upper arm with respect to the trunk in the sagittal (side-view) plane. It captures flexion and extension of the shoulder:

$$\alpha_s = \arccos\left(\frac{(\mathbf{r}_t - \mathbf{r}_n)^\top (\mathbf{r}_e - \mathbf{r}_s)}{\|\mathbf{r}_t - \mathbf{r}_n\| \|\mathbf{r}_e - \mathbf{r}_s\|}\right) \quad (15)$$

Here, $\mathbf{r}_t - \mathbf{r}_n$ defines the trunk direction and $\mathbf{r}_e - \mathbf{r}_s$ defines the upper arm direction. When the arm hangs naturally alongside the body, $\alpha_s \approx 0^\circ$; high values indicate overhead reaching, which sharply increases shoulder load.

The **upper arm coronal angle** α_c quantifies the lateral abduction of the upper arm relative to the frontal plane of the torso:

$$\alpha_c = \arccos\left(\frac{(\mathbf{r}_n - \mathbf{r}_s)^\top (\mathbf{r}_e - \mathbf{r}_s)}{\|\mathbf{r}_n - \mathbf{r}_s\| \|\mathbf{r}_e - \mathbf{r}_s\|}\right) - 90^\circ \quad (16)$$

The 90° offset re-centers the angle so that $\alpha_c = 0^\circ$ corresponds to the arm aligned with the coronal plane, and positive values indicate abduction away from the body.

The **lower arm sagittal angle** β_s describes the flexion of the elbow:

$$\beta_s = \arccos\left(\frac{(\mathbf{r}_w - \mathbf{r}_e)^\top (\mathbf{r}_i - \mathbf{r}_s)}{\|\mathbf{r}_w - \mathbf{r}_e\| \|\mathbf{r}_i - \mathbf{r}_s\|}\right) \quad (17)$$

A fully extended elbow corresponds to $\beta_s \approx 180^\circ$, while acute values indicate deep flexion. RULA assigns increasing scores for elbow angles that deviate far from the 60° – 100° comfort range.

The **forearm transversal deviation** β_t captures whether the forearm crosses the body midline, a posture that increases pronation stress:

$$\beta_t = 90^\circ - \arccos\left(\frac{(\mathbf{r}_s - \mathbf{r}_n)^\top (\mathbf{r}_w - \mathbf{r}_e)}{\|\mathbf{r}_s - \mathbf{r}_n\| \|\mathbf{r}_w - \mathbf{r}_e\|}\right) \quad (18)$$

Positive β_t indicates that the forearm deviates laterally beyond the shoulder line, while negative values indicate midline crossing.

Finally, the **wrist bend** γ_s measures the angular deviation between the forearm and the hand, capturing wrist flexion and extension:

$$\gamma_s = \arccos\left(\frac{(\mathbf{r}_w - \mathbf{r}_e)^\top (\mathbf{r}_i - \mathbf{r}_w)}{\|\mathbf{r}_w - \mathbf{r}_e\| \|\mathbf{r}_i - \mathbf{r}_w\|}\right) \quad (19)$$

A neutral (straight) wrist corresponds to $\gamma_s \approx 180^\circ$; deviations indicate flexion or extension, both of which increase the risk of carpal tunnel stress during sustained load carrying.

Together, these five angles constitute the ergonomic state vector $\Theta_{arm}(t)$, which is passed to the Arm Comfort Score module described next. Because all computations rely solely on 3D joint positions already available from the vision pipeline, no additional sensors are required.

C. Adaptive Weight Generator

Having established both a predictive model of future hand motion (Section III-A) and a biomechanical assessment of current arm posture (Section III-B), the system must combine these two independent evaluations into a single, actionable control signal. The Adaptive Weight Generator fulfills this role by producing a scalar blending weight $w(t) \in [0, 1]$ at each control tick, where $w \rightarrow 1$ grants full authority to the Deep-GRU predicted trajectory and $w \rightarrow 0$ reverts control entirely to the robot's own feedback-based trajectory. The weight is synthesized from two expert scores—prediction reliability and ergonomic comfort—using a Product-of-Experts fusion mechanism with a Lyapunov-stable update law.

1) *Module A: Prediction Reliability Score*: The raw prediction error $\epsilon(t)$ from (11) may exhibit large transient spikes caused by momentary sensor noise or brief occlusions, which do not reflect a genuine degradation of the prediction model. Reacting to such spikes would cause unnecessary authority oscillations and degrade the smoothness of robot behavior. To suppress these transients while still tracking genuine trends in prediction quality, $\epsilon(t)$ is filtered using an Exponential Moving Average (EMA):

$$\bar{\epsilon}(t) = \alpha_\epsilon \bar{\epsilon}(t-1) + (1 - \alpha_\epsilon) \epsilon(t), \quad \alpha_\epsilon \in (0, 1) \quad (20)$$

where α_ϵ is the smoothing coefficient. A higher α_ϵ yields a longer effective memory, making the filter more robust to short-lived spikes but slower to respond to sustained error increases. In practice, α_ϵ is selected so that the effective window length $N_{\text{eff}} \approx 1/(1 - \alpha_\epsilon)$ spans a few control cycles, balancing noise rejection with responsiveness.

The smoothed error $\bar{\epsilon}(t)$ must next be converted into a bounded trust metric. A natural choice is an exponential decay function, which maps zero error to perfect trust ($s_r = 1$) and

monotonically penalizes increasing errors with a decay rate that can be physically tuned:

$$s_r(t) = \exp(-\lambda \bar{e}(t)), \quad \lambda > 0 \quad (21)$$

The parameter λ governs the sensitivity of the trust score to prediction degradation. Specifically, λ can be interpreted as the inverse of the error half-life: the smoothed error at which the reliability drops to $1/e \approx 0.37$ is $\bar{e} = 1/\lambda$. A large λ makes the system conservative, rapidly withdrawing trust for even moderate prediction deviations; a small λ makes it tolerant, maintaining high trust unless errors become substantial.

2) *Module B: Arm Comfort Score*: The second expert score quantifies the ergonomic quality of the operator's current arm posture relative to an ideal, low-risk configuration. Because the five joint angles in $\Theta_{arm}(t)$ span different physiological ranges—for instance, shoulder flexion may vary over $[0^\circ, 180^\circ]$ while wrist bend may only span $[150^\circ, 210^\circ]$ —a simple Euclidean distance from the optimal posture would be dominated by whichever angle has the largest absolute range. To properly account for these differing scales and for the statistical correlations between angles (e.g., shoulder flexion and elbow flexion are often coupled), the Mahalanobis distance [34] is employed. This metric normalizes the deviation of each angle by the variance observed in the optimal posture dataset and accounts for inter-joint covariances, yielding a scale-invariant measure of how far the current posture deviates from the optimal distribution.

Offline phase. An optimal pose dataset $\Theta_{opt} = \{\Theta_{opt}^{(i)}\}_{i=1}^M \subset \mathbb{R}^5$ is formed by sampling arm configurations that achieve the best possible RULA score (score 1), which corresponds to minimal musculoskeletal risk. The assumption underlying this approach is that the set of optimal postures can be adequately characterized by a multivariate Gaussian distribution, which is reasonable given that RULA score-1 configurations cluster within narrow angular bands around the natural resting posture. From this dataset, the sample mean vector $\mu_{opt} \in \mathbb{R}^{5 \times 1}$ and the regularized covariance matrix $\Sigma_{reg} \in \mathbb{R}^{5 \times 5}$ are computed as:

$$\mu_{opt} = \frac{1}{M} \sum_{i=1}^M \Theta_{opt}^{(i)} = [\bar{\alpha}_s, \bar{\alpha}_c, \bar{\beta}_s, \bar{\beta}_t, \bar{\gamma}_s]^\top \quad (22)$$

$$\Sigma_{opt} = \frac{1}{M-1} \sum_{i=1}^M (\Theta_{opt}^{(i)} - \mu_{opt})(\Theta_{opt}^{(i)} - \mu_{opt})^\top \quad (23)$$

$$\Sigma_{reg} = \Sigma_{opt} + \delta \mathbf{I}_5 = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{15} \\ \vdots & \ddots & \vdots \\ \sigma_{51} & \cdots & \sigma_{55} \end{bmatrix} + \delta \mathbf{I}_5 \quad (24)$$

where $\mathbf{I}_5$ is the 5×5 identity matrix, σ_{ij} represents the variance (when $i = j$) or covariance (when $i \neq j$) between the i -th and j -th joint angles, and $\delta > 0$ is a small regularization parameter. The regularization is necessary because certain optimal postures may exhibit near-zero variance along particular joint dimensions (e.g., the wrist remains nearly straight in all score-1 configurations), which would render Σ_{opt} singular. Adding $\delta \mathbf{I}_5$ ensures that the matrix remains positive-definite and invertible.

Online phase. At each control tick, the Mahalanobis distance from the current arm pose $\Theta_{arm}(t)$ to the optimal distribution is computed as:

$$d(t) = \sqrt{(\Theta_{arm}(t) - \mu_{opt})^\top \Sigma_{reg}^{-1} (\Theta_{arm}(t) - \mu_{opt})} \quad (25)$$

Geometrically, $d(t)$ measures the number of “standard deviations” by which the current posture deviates from the optimal mean, where each deviation is weighted by the inverse covariance to account for the natural variability and coupling of each joint angle. A small $d(t)$ indicates that the operator's arm configuration is close to an ergonomically ideal posture; a large $d(t)$ signals increasing musculoskeletal risk.

The Mahalanobis distance is then mapped to a comfort score via an exponential decay, analogous to the reliability score:

$$s_e(t) = \exp(-\kappa d(t)), \quad \kappa > 0 \quad (26)$$

where κ scales the distance penalty. The parameter κ determines how aggressively the system reduces robot authority as the operator deviates from optimal posture: a large κ causes the comfort score to drop sharply even for moderate deviations, while a small κ maintains high comfort scores unless the posture becomes severely compromised.

3) *Module C: Weight Fusion and Adaptive Law*: Given the two expert scores—the prediction reliability $s_r(t) \in [0, 1]$ and the arm comfort $s_e(t) \in [0, 1]$ —the system must fuse them into a single target blending weight. A simple arithmetic or geometric mean would allow a high score in one dimension to compensate for a low score in the other. In shared control for human safety, however, both conditions must be simultaneously satisfied: the robot should only follow the predicted trajectory aggressively when the prediction is reliable *and* the operator's posture is comfortable. This AND-logic is naturally realized by a multiplicative Product-of-Experts formulation:

Product-of-Experts target weight. The target blending weight is formulated using a Product-of-Experts structure:

$$w^*(t) = s_r(t)^p \cdot s_e(t)^q, \quad p, q > 0 \quad (27)$$

This multiplicative form enforces AND-logic, ensuring $w^* \rightarrow 0$ whenever prediction reliability or ergonomic comfort drops significantly. The exponents p and q adjust the relative influence of reliability and ergonomics: for instance, setting $p > q$ makes the system more sensitive to prediction degradation than to postural discomfort.

Lyapunov-stable adaptive update. Directly assigning $w(t) = w^*(t)$ at each time step would cause discontinuous jumps in the blending weight whenever either expert score changes abruptly, leading to jerky robot motion and potential safety hazards. To guarantee continuous and asymptotically stable weight transitions, a first-order adaptive law is derived from Lyapunov stability theory. Defining the Lyapunov candidate $V(t) = \frac{1}{2}(w(t) - w^*(t))^2 \geq 0$, which is positive definite in the tracking error $w - w^*$, continuous dynamics $\dot{w} = -\eta(w - w^*)$ ensure:

$$\dot{V} = (w - w^*)\dot{w} = -\eta(w - w^*)^2 \leq 0 \quad (28)$$

where $\eta > 0$ determines the adaptation rate. Since $\dot{V} \leq 0$ with equality only when $w = w^*$, the weight $w(t)$ converges

asymptotically to the target $w^*(t)$, provided that $w^*(t)$ varies slowly relative to the convergence rate η . In practice, this slow-variation assumption holds because both $s_r(t)$ and $s_e(t)$ are already smoothed by their respective EMA filters.

For digital implementation at control frequency f_c , the continuous adaptive law is discretized and augmented with a rate limiter to prevent excessively rapid authority transfers. The discretized, rate-limited update rule is expressed as:

$$\Delta w(t) = \delta_w (w^*(t) - w(t)), \quad \delta_w = \eta / f_c \quad (29)$$

$$\Delta w_{\text{clamp}}(t) = \text{clip}(\Delta w(t), -\Delta w_{\text{max}}, +\Delta w_{\text{max}}) \quad (30)$$

$$w(t+1) = \text{clip}(w(t) + \Delta w_{\text{clamp}}(t), 0, 1) \quad (31)$$

where f_c is the control loop frequency in Hertz, $\delta_w = \eta / f_c$ is the per-step gain, and Δw_{max} limits the maximum per-step rate of authority change, acting as a saturation constraint that further prevents abrupt transitions even when the target weight changes rapidly.

Safety override with hysteresis. As a final safeguard, a hysteretic comparator overrides the target weight to immediately revoke predictive authority when prediction reliability falls below a critical threshold:

$$w^*(t) \leftarrow 0 \text{ if } \text{safety_mode} = \text{true} \quad (32)$$

Safety mode is triggered when $s_r(t) < s_{r,\text{crit}}$ and cleared when $s_r(t) > s_{r,\text{crit}} + \Delta_h$, where $s_{r,\text{crit}}$ is the critical reliability threshold and Δ_h is the hysteresis deadband. The hysteresis prevents rapid toggling (chattering) between normal and safety modes when s_r fluctuates around the threshold.

D. Trajectory Smoothing and Shared Control

The blending weight $w(t)$ produced by the Adaptive Weight Generator determines the balance between the predicted human trajectory and the robot's own feedback-based position. However, even though the weight itself transitions smoothly thanks to the Lyapunov adaptive law, the predicted position $\hat{\mathbf{p}}_{\text{pre}}(t+k)$ may contain high-frequency jitter originating from the vision pipeline (e.g., landmark detection noise, depth sensor quantization). Sending such noisy Cartesian commands directly to the robot controller would excite mechanical vibrations and compromise both tracking accuracy and operator comfort. The Trajectory Smoother addresses this by applying spatial filtering and velocity saturation to the blended trajectory.

The raw blended position at each control tick combines the predicted and feedback trajectories according to the current weight:

$$\mathbf{p}_{\text{raw}}(t) = w(t) \hat{\mathbf{p}}_{\text{pre}}(t+k) + (1-w(t)) \mathbf{p}_{fb}(t) \quad (33)$$

where $\mathbf{p}_{fb}(t)$ is the current end-effector position obtained from the robot's forward kinematics. When $w = 1$, the robot fully tracks the predicted trajectory; when $w = 0$, it holds its current position based on proprioceptive feedback.

To attenuate the high-frequency spatial noise present in $\mathbf{p}_{\text{raw}}(t)$, it is passed through an EMA low-pass filter:

$$\mathbf{p}_{sm}(t) = (1 - \alpha_s) \mathbf{p}_{sm}(t-1) + \alpha_s \mathbf{p}_{\text{raw}}(t), \quad \alpha_s \in (0, 1] \quad (34)$$

where α_s is the trajectory smoothing factor. A small α_s produces a heavier filter with greater noise attenuation but introduces more tracking lag; a large α_s yields faster response at the cost of less smoothing.

Even after filtering, the resulting trajectory may demand Cartesian velocities that exceed the physical capabilities of the robot or violate safety constraints. To enforce a velocity bound, the instantaneous end-effector velocity is computed as:

$$\mathbf{v}(t) = [\mathbf{p}_{sm}(t) - \mathbf{p}_{sm}(t-1)] \cdot f_c \quad (35)$$

If the velocity magnitude exceeds the maximum allowable end-effector velocity v_{max} , i.e., $\|\mathbf{v}(t)\| > v_{\text{max}}$, the smoothed position is projected back onto the velocity-feasible region by rescaling the displacement:

$$\mathbf{p}_{sm}(t) \leftarrow \mathbf{p}_{sm}(t-1) + \frac{v_{\text{max}}}{\|\mathbf{v}(t)\|} [\mathbf{p}_{sm}(t) - \mathbf{p}_{sm}(t-1)] \quad (36)$$

This saturation preserves the direction of the commanded motion while limiting its magnitude, ensuring that the robot never attempts to move faster than v_{max} regardless of how rapidly the blended trajectory changes.

E. Robust Robot Control and Kinematics

The preceding modules produce a smooth, velocity-bounded Cartesian trajectory $\mathbf{p}_{sm}(t)$ that encodes the desired end-effector position. To execute this trajectory on a physical robot arm, two additional transformations are required: first, the Cartesian command must be mapped into the robot's joint space via inverse kinematics; second, a joint-level controller must generate the velocity and position commands that drive the actuators while maintaining stability and minimizing mechanical stress.

1) *Quaternion-based Inverse Kinematics:* The inverse kinematics (IK) module converts the smoothed Cartesian command $\mathbf{p}_{sm}(t)$ to a joint reference $\mathbf{q}_{ref}(t) \in \mathbb{R}^6$. In addition to translational tracking, the end-effector must maintain a prescribed orientation during co-carrying to keep the grasped object stable. The target orientation $\mathbf{R}_{target} \in SO(3)$ is specified as a constant rotation matrix corresponding to the nominal grasp orientation and remains fixed throughout the task. The combined position and orientation error is computed using the $SO(3)$ logarithmic map:

$$\mathbf{e}(t) = \begin{bmatrix} \mathbf{p}_{sm}(t) - \mathbf{p}(q) \\ 0.5 \cdot \log_{SO(3)}(\mathbf{R}_{target} \mathbf{R}(q)^\top) \end{bmatrix} \in \mathbb{R}^6 \quad (37)$$

where $\mathbf{p}(q) \in \mathbb{R}^3$ and $\mathbf{R}(q) \in SO(3)$ are the current end-effector position and orientation obtained from forward kinematics at the current joint configuration q , and the logarithmic map extracts the axis-angle representation of the orientation error. The factor 0.5 provides a scaling that yields well-conditioned convergence behavior when combined with the translational error.

To resolve this 6-DoF error into a joint update, the Damped Least Squares (DLS) method [35] is employed, which prevents the numerical instabilities that arise near kinematic singularities in the standard pseudo-inverse approach. Letting

$\mathbf{J} \in \mathbb{R}^{6 \times 6}$ denote the geometric Jacobian of the robot at the current configuration, the differential joint update is:

$$\Delta q = \mathbf{J}^\top (\mathbf{J}\mathbf{J}^\top + \lambda_{DLS}^2 \mathbf{I}_6)^{-1} \mathbf{e} \quad (38)$$

where $\lambda_{DLS} > 0$ is the damping factor and $\mathbf{I}_6$ is the 6×6 identity matrix. The damping term $\lambda_{DLS}^2 \mathbf{I}_6$ regularizes the inversion, trading off tracking accuracy for numerical stability: near singularities where $\mathbf{J}\mathbf{J}^\top$ becomes ill-conditioned, the damping prevents the joint velocities from diverging. Joint configurations are then updated iteratively with a step size $\alpha_{IK} \in (0, 1]$ and clamped to respect the physical joint limits of the robot:

$$q \leftarrow \text{clip}(q + \alpha_{IK} \Delta q, \mathbf{q}_{\min}, \mathbf{q}_{\max}) \quad (39)$$

where $\mathbf{q}_{\min}$ and $\mathbf{q}_{\max}$ are the vectors of lower and upper joint limits, respectively.

2) *LQR Velocity Controller*: The IK solver produces a joint reference $\mathbf{q}_{ref}(t)$, but sending this reference directly to the actuators as a position command would result in abrupt, high-acceleration motions that stress the mechanical components and compromise operator comfort. Instead, a Linear-Quadratic Regulator (LQR) is designed to compute optimal joint velocity commands that drive each joint toward its reference smoothly, explicitly balancing tracking performance against actuator effort.

To formulate the LQR, each of the robot's six joints is modeled independently as a double integrator, which captures the essential relationship between position, velocity, and acceleration while remaining computationally tractable for real-time control. For the i -th joint, the state vector, control input, and continuous-time dynamics are:

$$\mathbf{x}_i = \begin{bmatrix} q_i - q_{ref,i} \\ \dot{q}_i \end{bmatrix}, \quad u_i = \ddot{q}_i, \quad \dot{\mathbf{x}}_i = \mathbf{A}\mathbf{x}_i + \mathbf{B}u_i \quad (40)$$

where the state $\mathbf{x}_i \in \mathbb{R}^2$ comprises the position error $q_i - q_{ref,i}$ and the current velocity $\dot{q}_i$, and the control input u_i is the commanded acceleration. The system matrices are $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, reflecting the kinematic chain from acceleration to velocity to position. Discretized with sample period $\Delta t = 1/f_c$ (where f_c is the control loop frequency), the discrete state matrices become $\mathbf{A}_d = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}$ and $\mathbf{B}_d = \begin{bmatrix} \Delta t^2/2 \\ \Delta t \end{bmatrix}$.

The optimal control law is obtained by minimizing a discrete infinite-horizon quadratic cost function that penalizes both sustained tracking errors and excessive control effort:

$$J_i = \sum_{k=0}^{\infty} [\mathbf{x}_i(k)^\top \mathbf{Q} \mathbf{x}_i(k) + u_i(k)^\top R u_i(k)] \quad (41)$$

where $\mathbf{Q} = \text{diag}(q_1, q_2) \in \mathbb{R}^{2 \times 2}$ is the state weighting matrix, with q_1 penalizing position error and q_2 penalizing velocity magnitude, and $R = \rho \in \mathbb{R}$ is the control weighting scalar penalizing acceleration magnitude. Increasing q_1 relative to ρ yields tighter tracking at the expense of more aggressive actuator usage; increasing ρ produces smoother, gentler motions.

The solution to this optimization is well-known from optimal control theory. The unique positive-definite symmetric

matrix $\mathbf{P} \in \mathbb{R}^{2 \times 2}$ satisfying the Discrete-time Algebraic Riccati Equation (DARE) encodes the optimal cost-to-go:

$$\mathbf{P} = \mathbf{A}_d^\top \mathbf{P} \mathbf{A}_d - (\mathbf{A}_d^\top \mathbf{P} \mathbf{B}_d)(R + \mathbf{B}_d^\top \mathbf{P} \mathbf{B}_d)^{-1} (\mathbf{B}_d^\top \mathbf{P} \mathbf{A}_d) + \mathbf{Q} \quad (42)$$

Solving the DARE yields the optimal state-feedback gain matrix $\mathbf{K} = [K_p, K_d] \in \mathbb{R}^{1 \times 2}$:

$$\mathbf{K} = (R + \mathbf{B}_d^\top \mathbf{P} \mathbf{B}_d)^{-1} \mathbf{B}_d^\top \mathbf{P} \mathbf{A}_d \quad (43)$$

The two components of $\mathbf{K}$ have a clear physical interpretation: K_p acts as a proportional gain on the position error, pulling the joint toward its reference, while K_d acts as a derivative (damping) gain on the velocity, preventing overshoot and oscillation. The resulting control law and the joint velocity and position commands sent to the robot hardware are:

$$u_i(t) = K_p(q_{ref,i} - q_{fb,i}) + K_d(0 - \dot{q}_{fb,i}) \quad (44)$$

$$\dot{q}_{cmd,i}(t) = \text{clip}(\dot{q}_{fb,i} + u_i \Delta t, \pm \dot{q}_{\max,i}) \quad (45)$$

$$q_{cmd,i}(t) = \text{clip}(q_{fb,i} + \dot{q}_{cmd,i} \Delta t, q_{s,\min,i}, q_{s,\max,i}) \quad (46)$$

where $q_{fb,i}$ and $\dot{q}_{fb,i}$ are the measured joint position and velocity from the robot's state publisher, and $\dot{q}_{\max,i}$, $q_{s,\min,i}$, $q_{s,\max,i}$ are the velocity and position safety limits for the i -th joint, respectively. The clipping operations enforce hard actuator constraints, ensuring that the optimal control signal is always physically realizable.

IV. EXPERIMENTAL SETUP

The proposed architecture was validated using a high-fidelity setup featuring an AI-capable camera for skeleton tracking and a physical collaborative robot arm. The software backend was developed using cross-platform C/C++ build systems managed via CMake to ensure real-time execution.

V. RESULTS AND DISCUSSION

To evaluate the effectiveness of the proposed framework, ablation studies were conducted against two baselines:

- 1) Fixed weight blending ($w = \text{const}$).
- 2) Reliability-only blending (ignoring comfort score s_j).

Results indicate that the proposed adaptive weight generator significantly reduces the root-mean-square error (RMSE) of the trajectory tracking while minimizing sharp torque gradients at the robot joints. Furthermore, analysis of the operator's posture demonstrated a measurable improvement in overall arm comfort over extended operational periods.

VI. CONCLUSION

This paper presented an adaptive shared control framework for human-robot collaboration that uniquely fuses deep learning prediction reliability with human ergonomic comfort. By ensuring mathematically stable authority blending and enforcing strict quaternion-based kinematic constraints, the proposed system yields safer, more fluent collaborative performance. Future work will investigate the integration of haptic feedback and the exploration of multi-agent collaborative environments.

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